AI Predictive Analytics for Optimised Crop Yield

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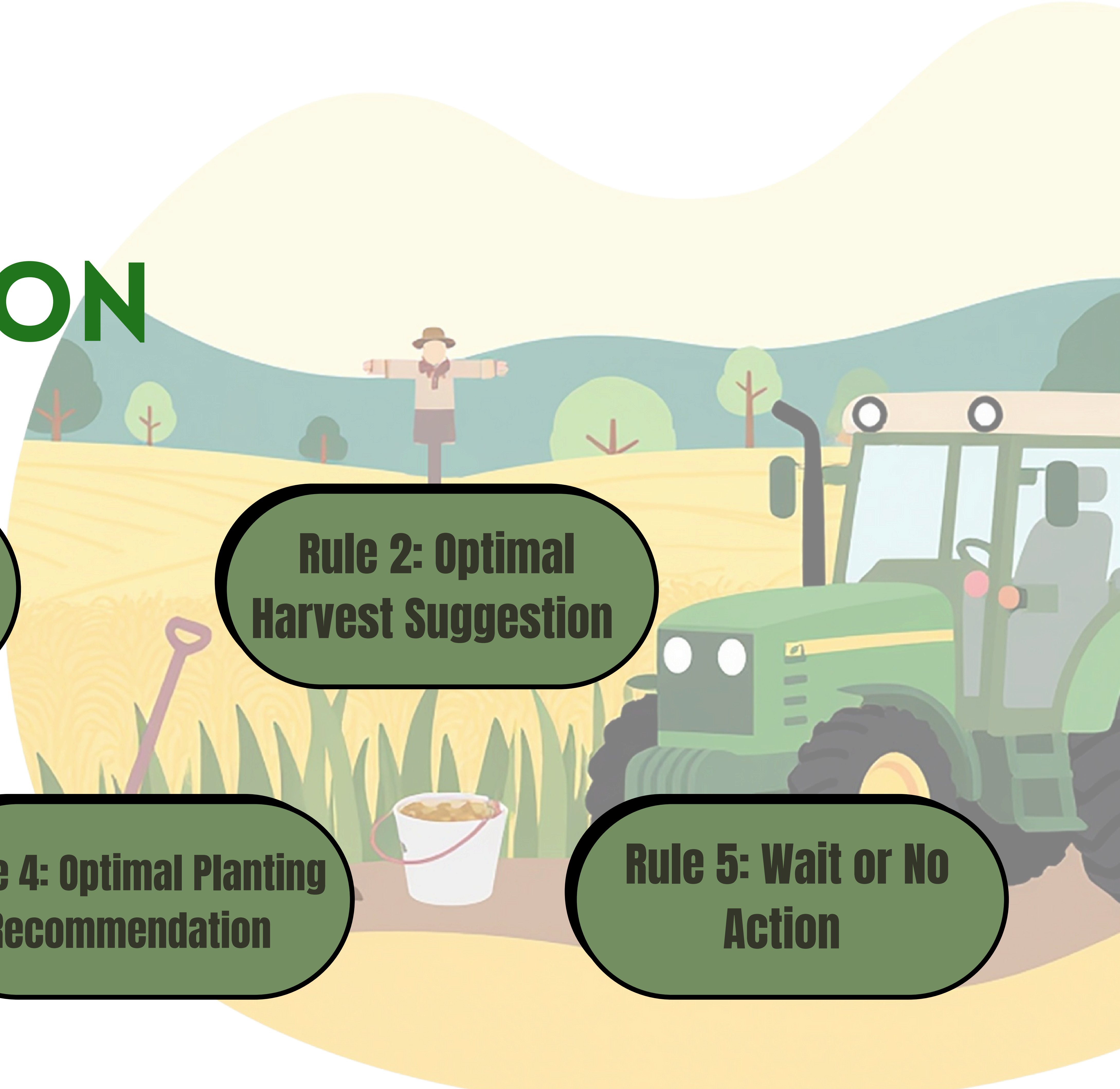
PROJECT PRESENTATION

INTRODUCTION

Unpredictable weather often disrupt planting and harvesting, causing crop damage, low yield, and **financial loss for farmers.** This project proposes a Smart Agriculture Forecasting System that uses AI to analyze weather, soil humidity, and crop data to predict the optimal planting time, the best harvesting period, and expected yield. The system helps farmers make data-driven decisions, reduce crop loss, improve productivity, and promote sustainable agricultural management through clear predictions and an interactive dashboard.

KNOWLEDGE REPRESENTATION



Rule 1: Emergency Harvest Suggestion

Rule 2: Optimal Harvest Suggestion

Rule 3: Predicted Yield Calculation

Rule 4: Optimal Planting Recommendation

Rule 5: Wait or No Action

Rule 1: Emergency Harvest Suggestion

Production Rules:

IF a crop is of a harvestable age, **AND** there is a high flood risk forecast within the next 3 days, **THEN** the agent must suggest an 'Early Harvest' advice to save the crop.

Predicate Logic/FOL:

$$\forall \mathrm{crop}[\mathrm{HarvestableAge}(\mathrm{crop}) \wedge \mathrm{FloodRiskHigh}() \rightarrow \mathrm{SuggestEarlyCrop}()]$$

Rule 2: Optimal Harvest Suggestion

Production Rules:

IF a crop's age is in the ‘Optimal Harvest’ for its type, **AND** the weather forecast shows 'Low' risk, **THEN** the agent will suggest an ‘Optimal Harvest’.

Predicate Logic/FOL:

$$\forall \mathrm{crop}[\mathrm{OptimalAge}(\mathrm{crop}) \land \mathrm{WeatherRiskLow}() \rightarrow \mathrm{SuggestOptimalHarvest}()]$$

Rule 3: Predicted Yield Calculation

Production Rules:

IF the agent is recommending any harvest action, either 'Early Harvest' or 'Optimal Harvest', **THEN** it must also check the 'Yield Model' to predict the expected yield for that crop at its current age.

Predicate Logic/FOL:

$$\forall\, crop[\text{SuggestEarlyHarvest}(crop) \lor \text{SuggestOptimalHarvest}(crop) \rightarrow \text{CheckYieldModel}(crop)]$$

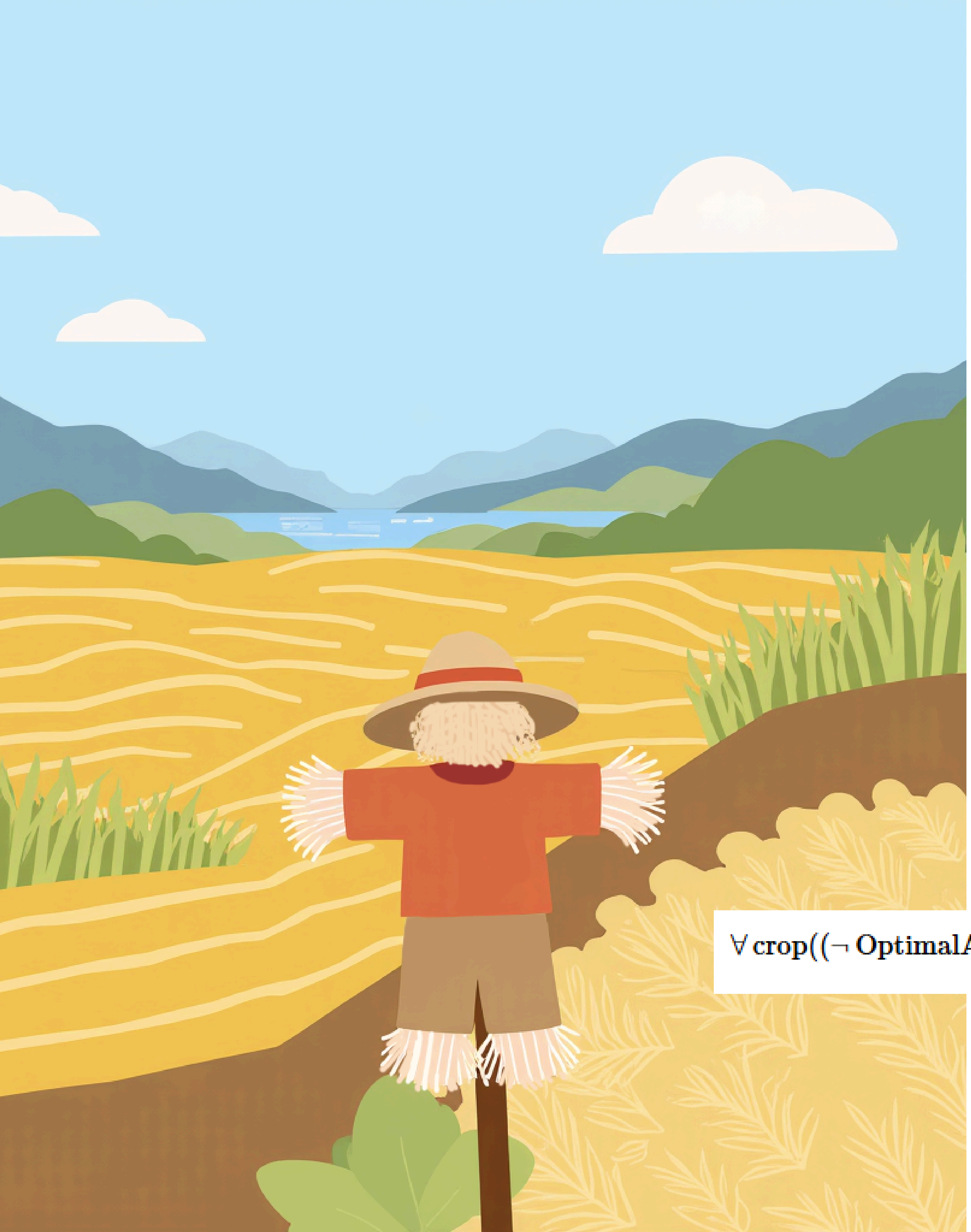
Rule 4: Optimal Planting Recommendation

Production Rules:

IF a farmer's plot is ‘Empty’, **AND** the humidity forecast for the next 2 days is in the ‘Optimal Planting’ for the crop they want, **THEN** the agent will suggest ‘Plant Now’.

Predicate Logic/FOL:

$$\forall \mathrm{plot},\ \mathrm{crop}[\mathrm{PlotEmpty}(\mathrm{crop}) \wedge \mathrm{HumidityOptimal}(\mathrm{crop}) \rightarrow \mathrm{SuggestPlanNow}(\mathrm{crop})]$$



Rule 5: Wait or No Action

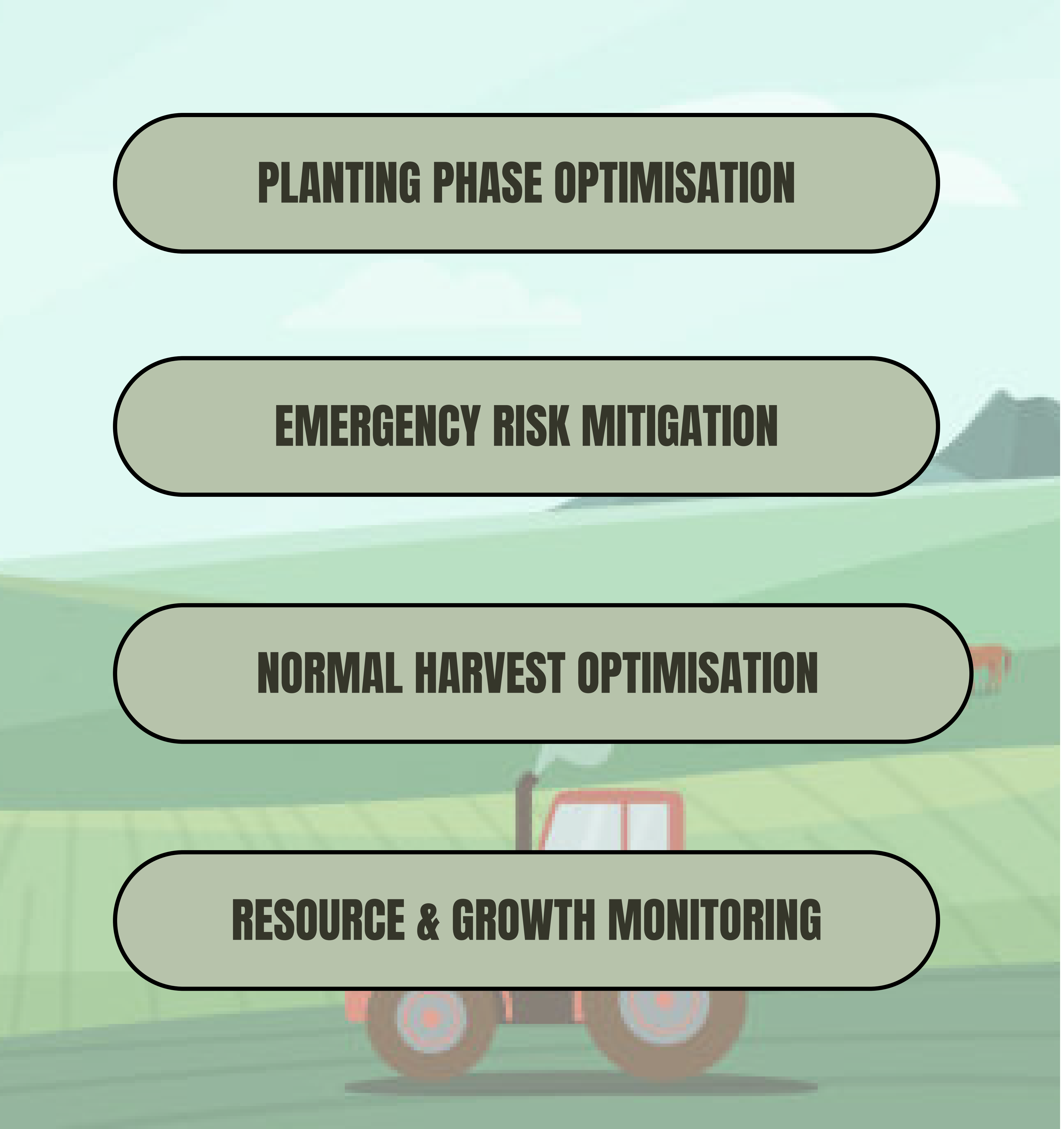
Production Rules:

IF a crop is not in the best harvest time**, AND** it is not old enough for an emergency harvest, **OR** the weather is not 'Low' risk for harvesting, **THEN** the agent will recommend to 'Wait'.

Predicate Logic/FOL:

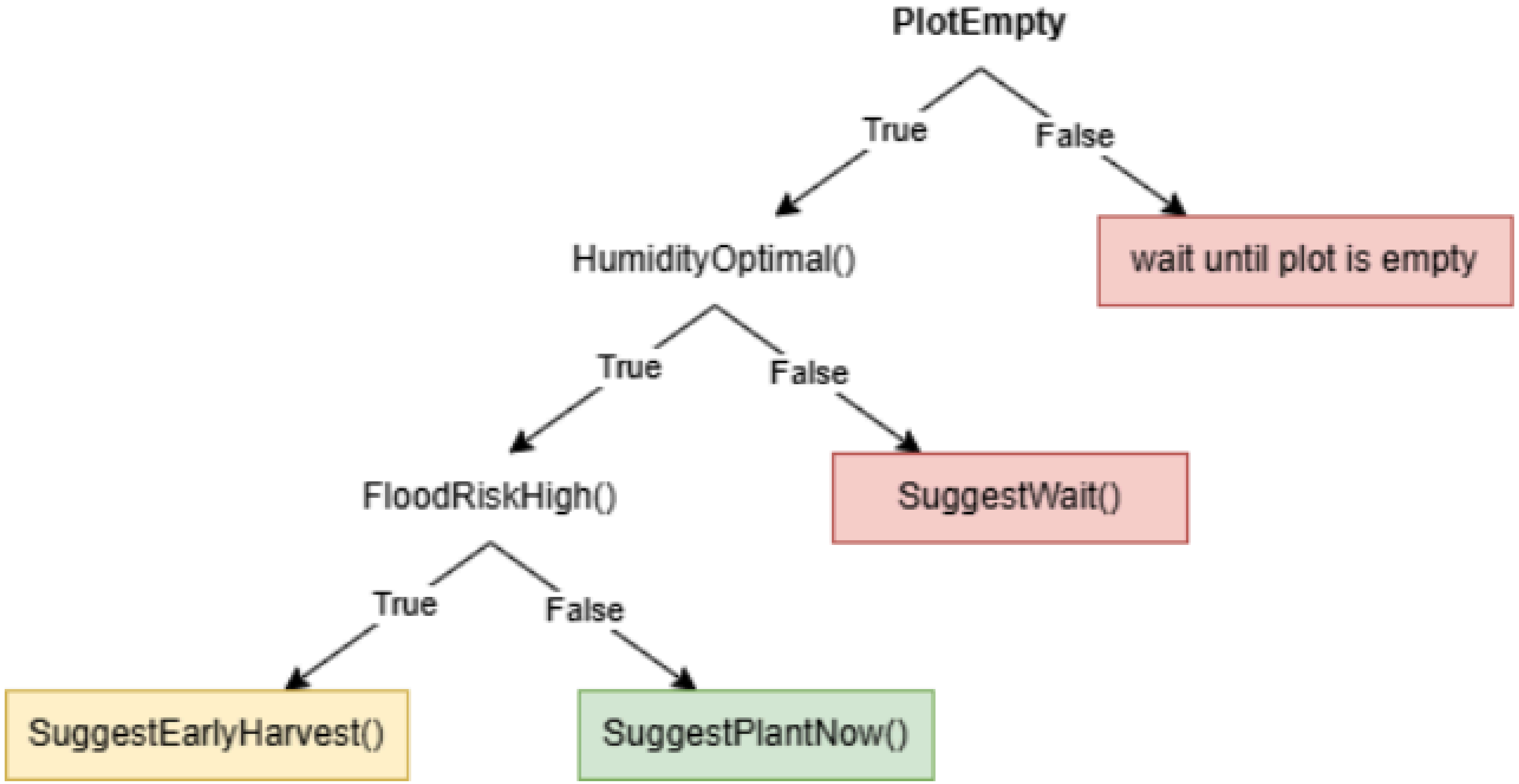
$$\forall \mathrm{crop}((\neg\, \mathrm{OptimalAge}(\mathrm{crop}) \land \neg\, \mathrm{HarvestableAge}(\mathrm{crop})) \lor \neg\, \mathrm{WeatherRiskLow}()) \rightarrow \mathrm{SuggestWait}(\mathrm{crop})]$$

STATE SPACE SEARCH

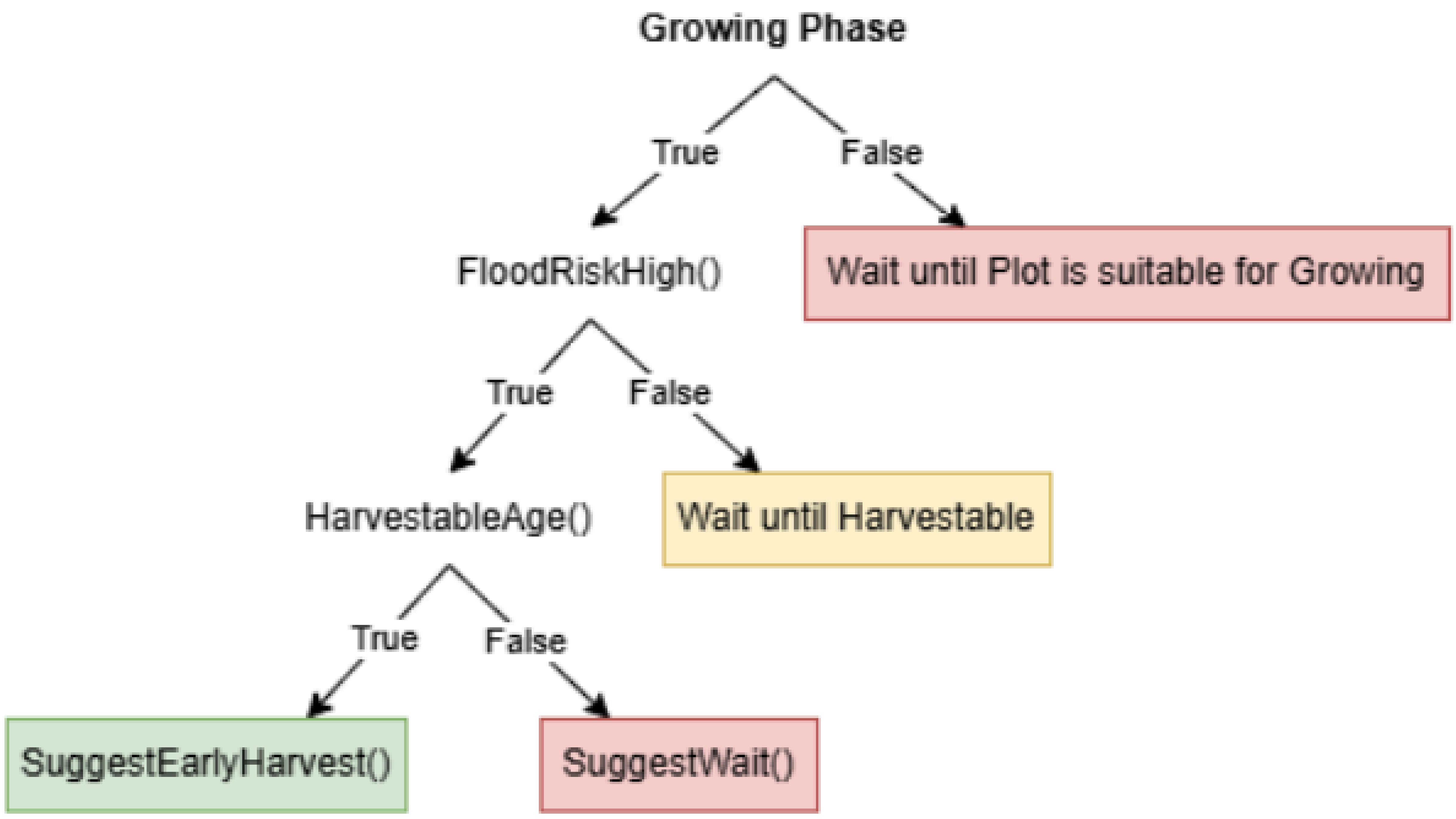


- PLANTING PHASE OPTIMISATION
- EMERGENCY RISK MITIGATION
- NORMAL HARVEST OPTIMISATION
- RESOURCE & GROWTH MONITORING

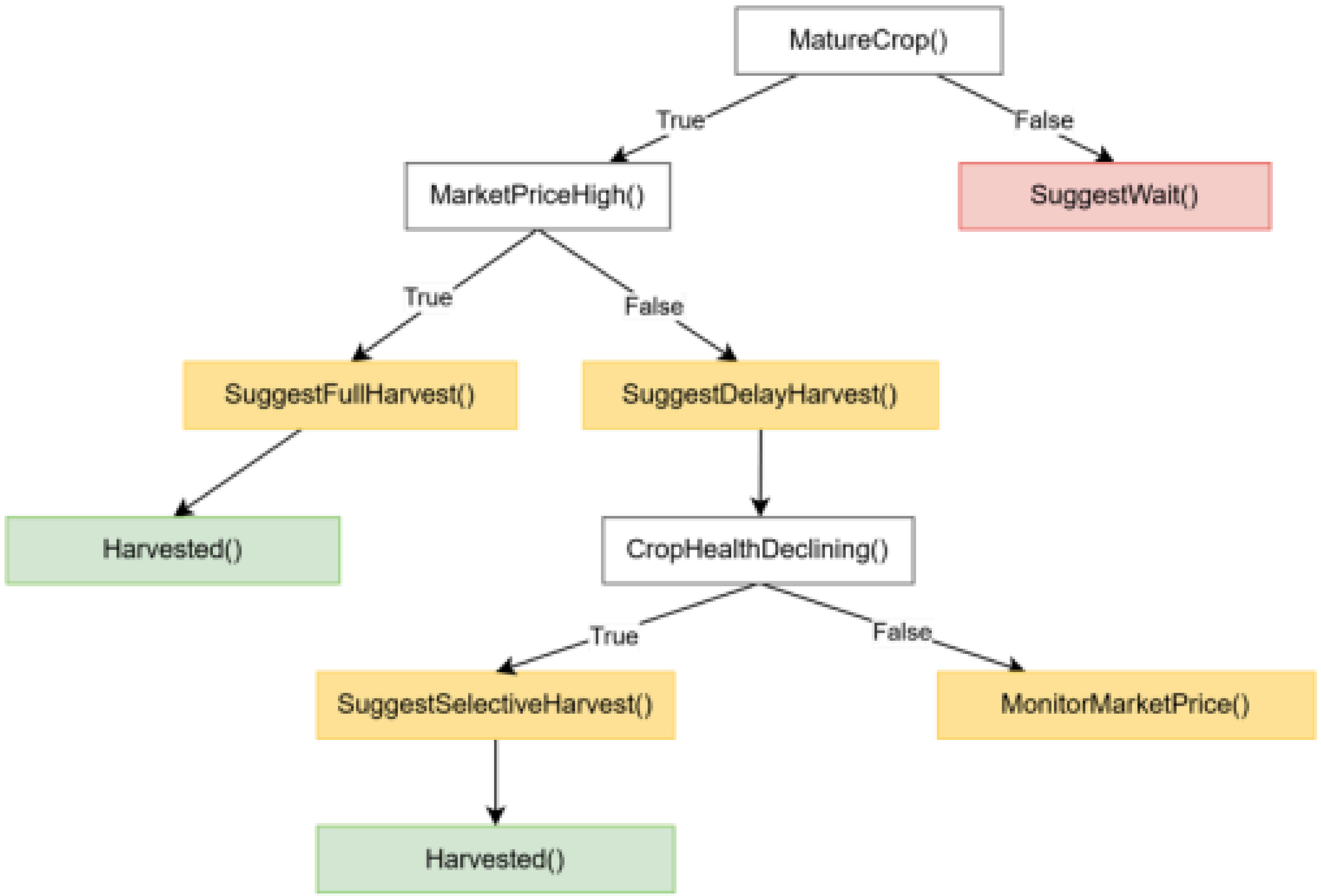
1.0: PLANTING PHASE OPTIMISATION



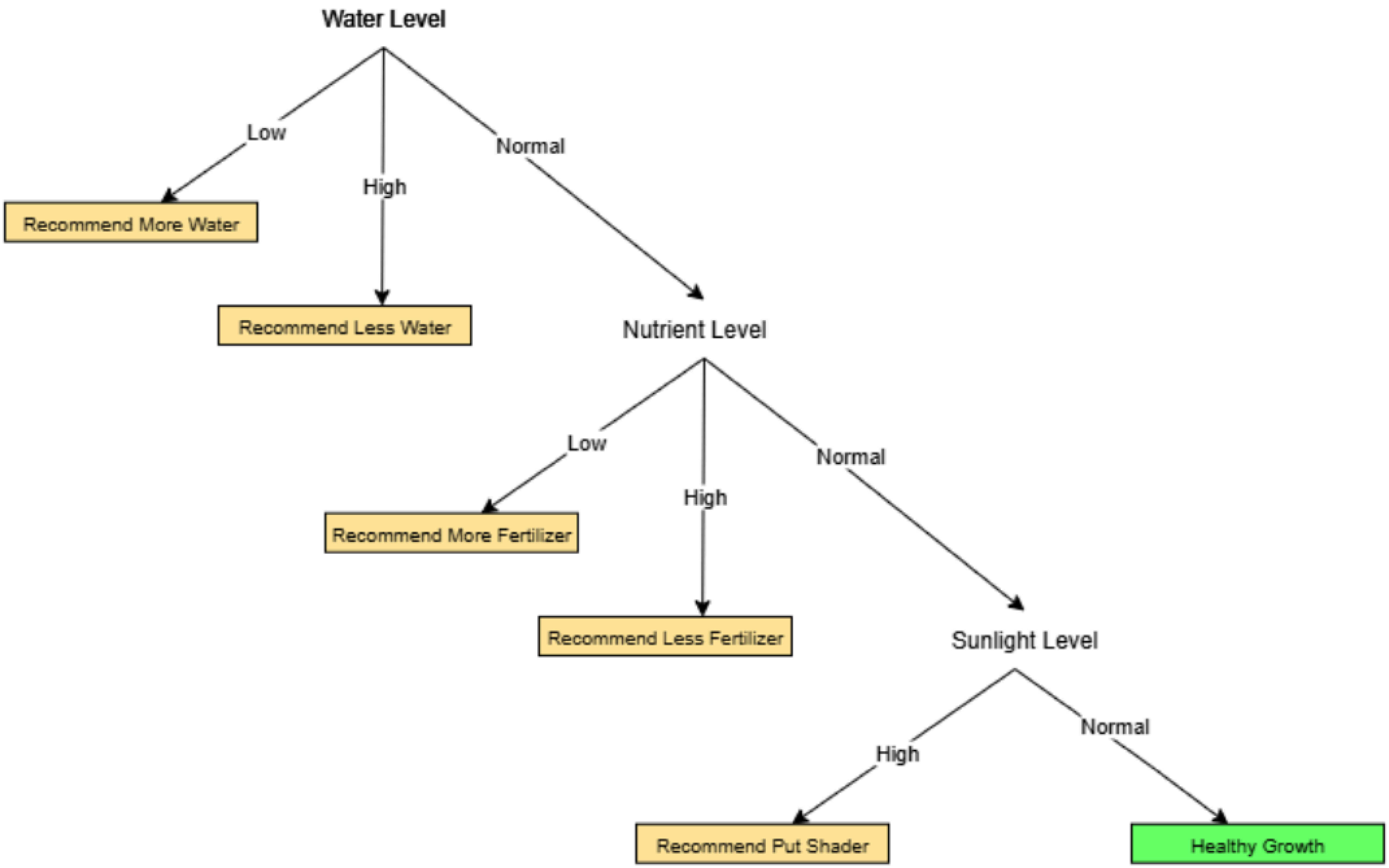
2.0: EMERGENCY RISK MITIGATION



3.0: NORMAL HARVEST OPTIMISATION

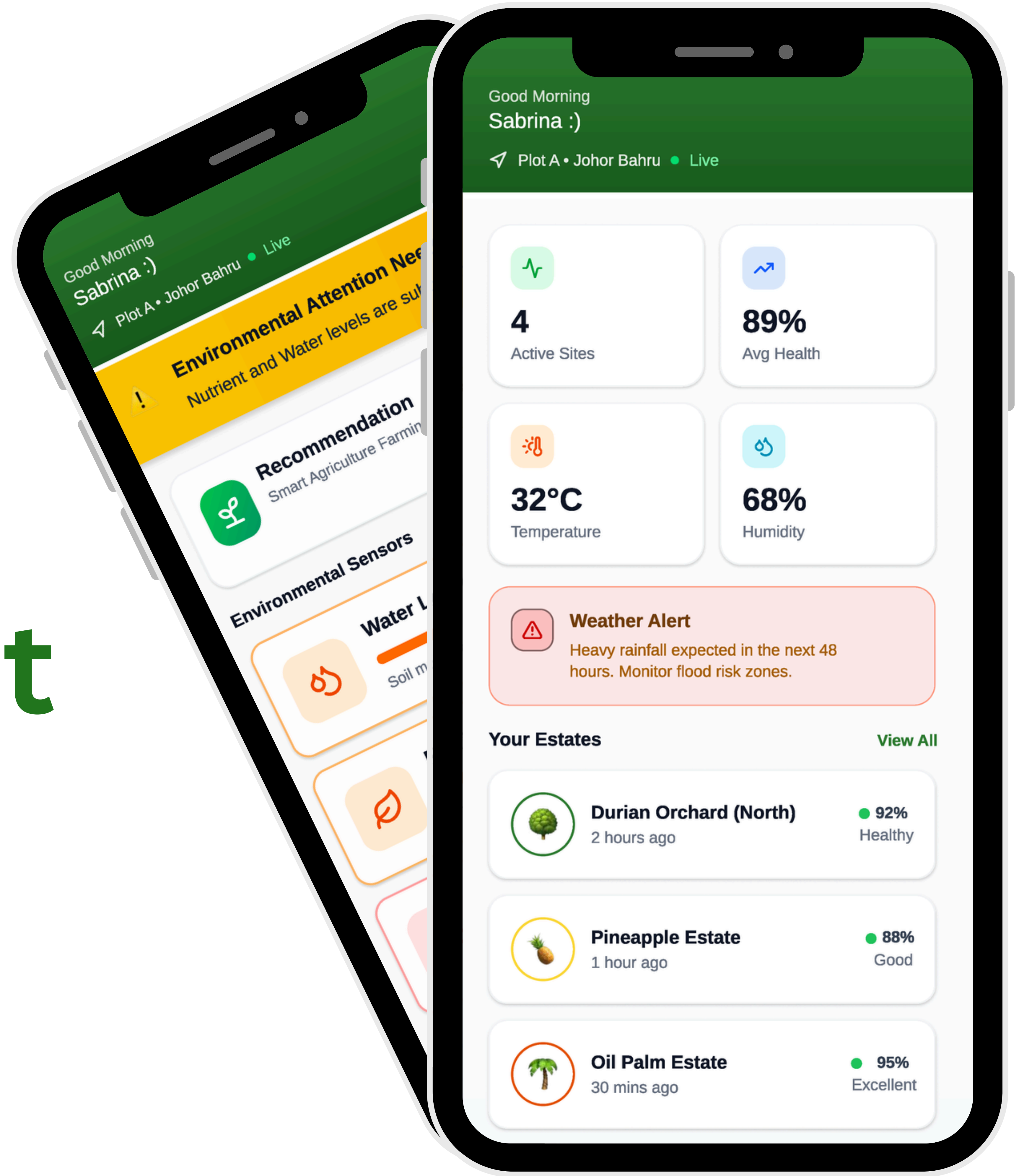


4.0: RESOURCE and GROWTH MONITORING

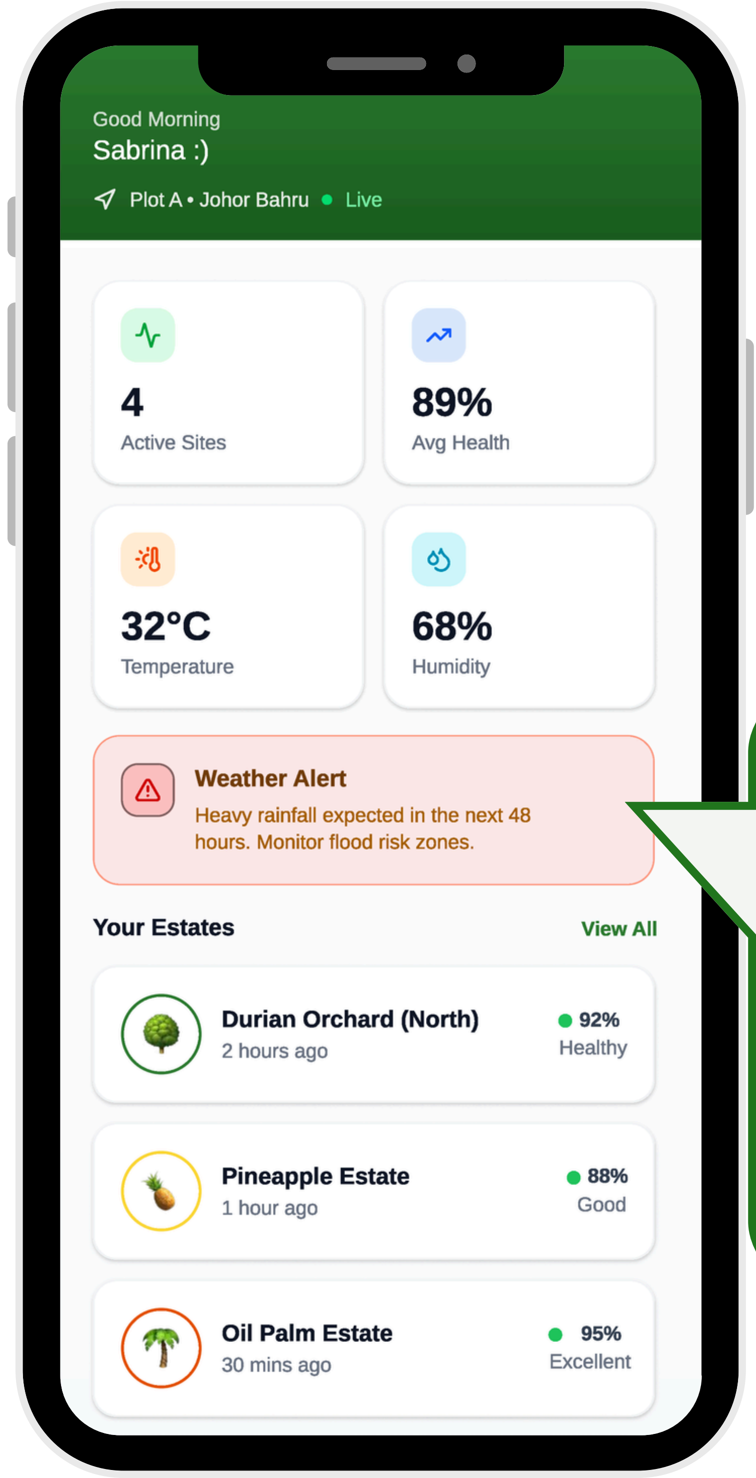


PEAS Model
Your paragraph text

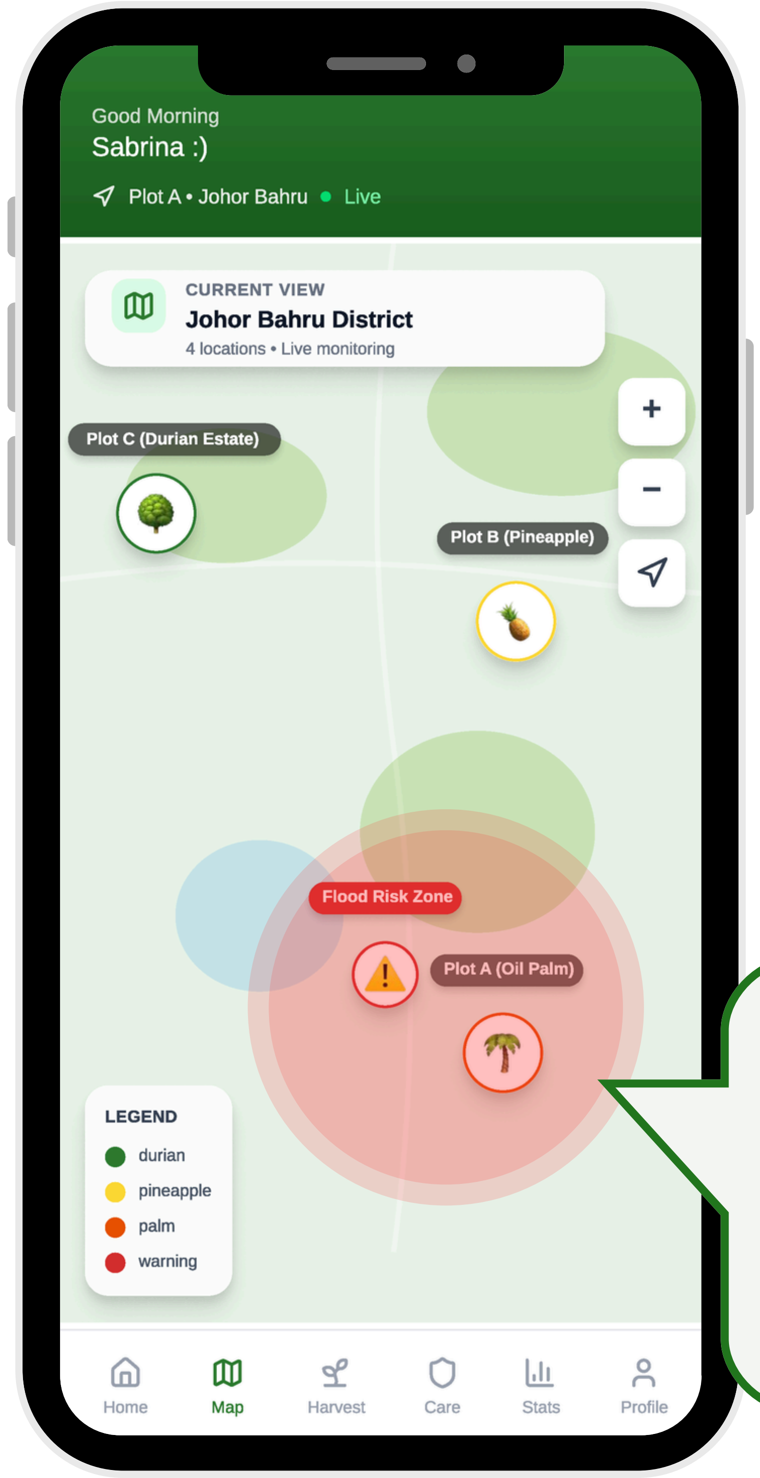
Proof of Concept Demonstration



Scenario 1: Critical Flood Risk Management



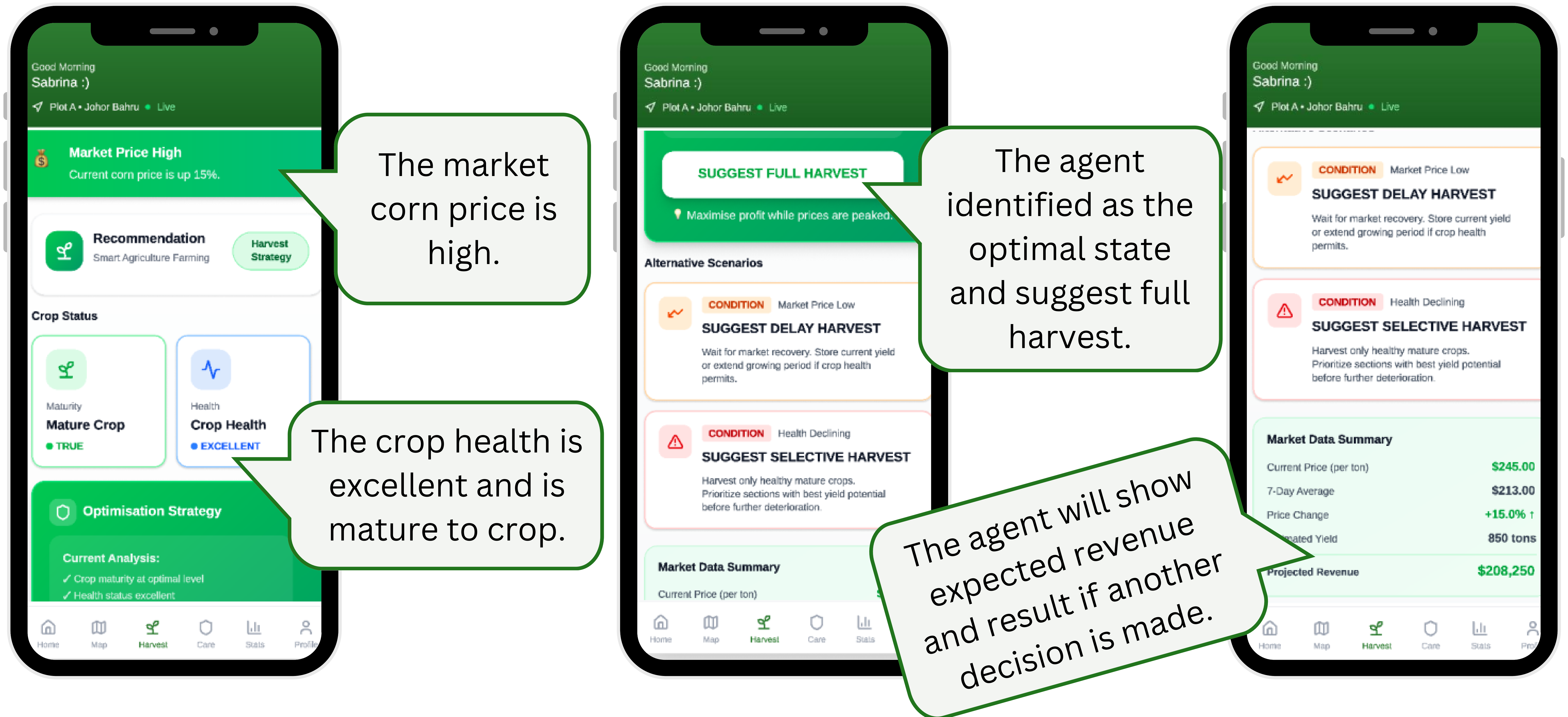
The weather API detect heavy rainfall in the next 48 hours.



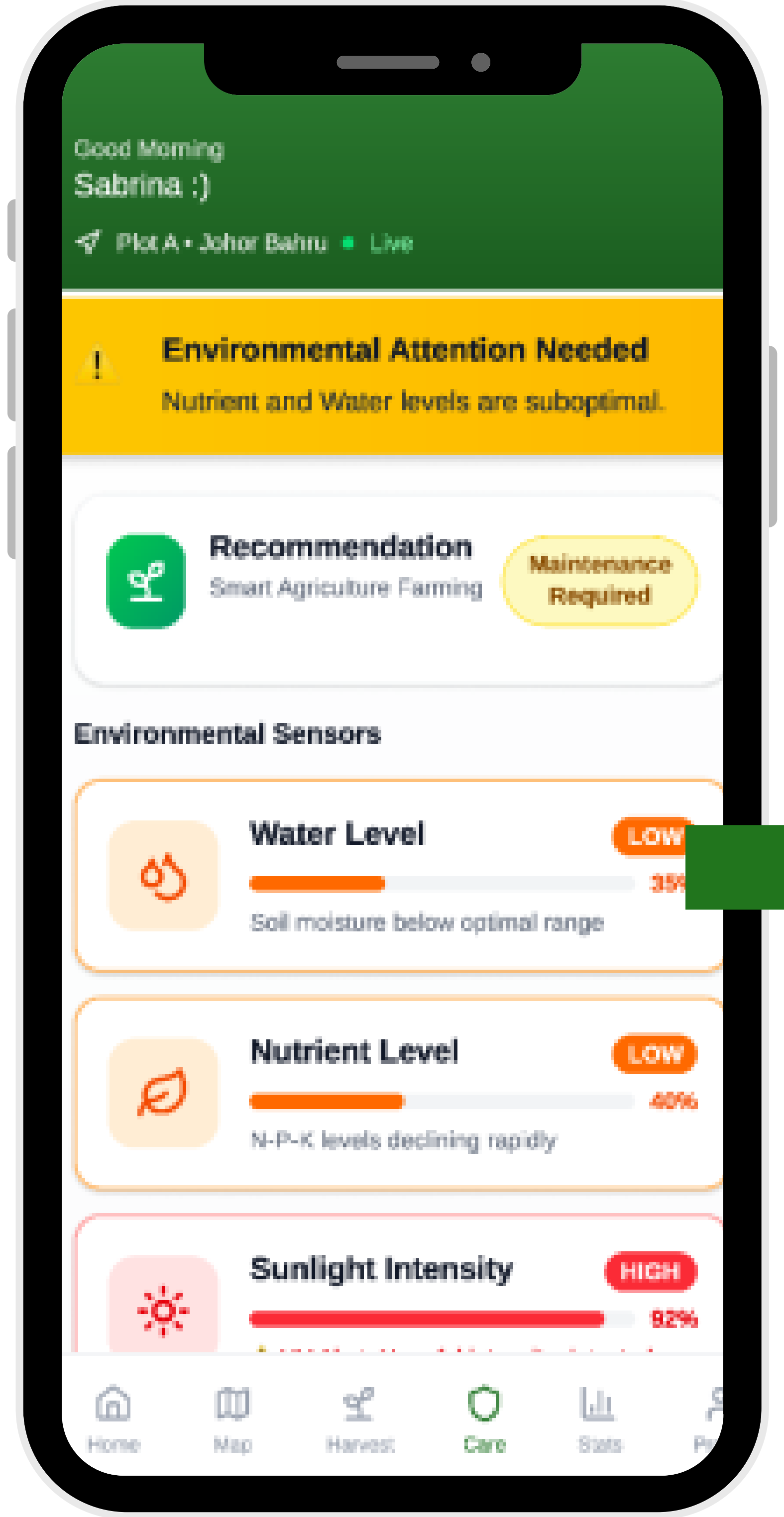
Red weather alert on dashboard map. Specific zone highlighted on map.

Rule *EarlyHarvest*() in the KB is triggered. Crop is at harvestable age.

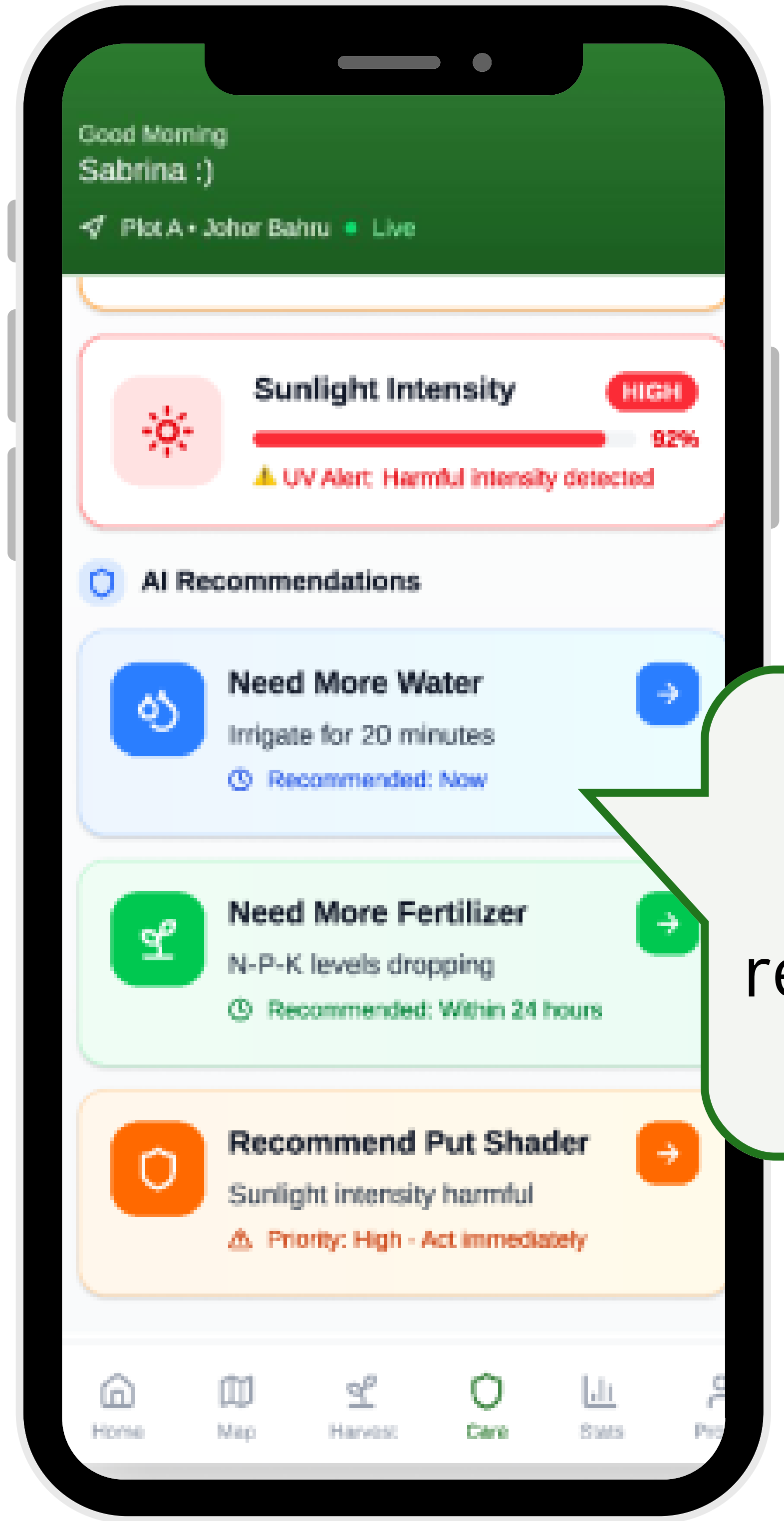
Scenario 2: Harvest Optimising Strategy



Scenario 3: Environmental Care & Recommendation

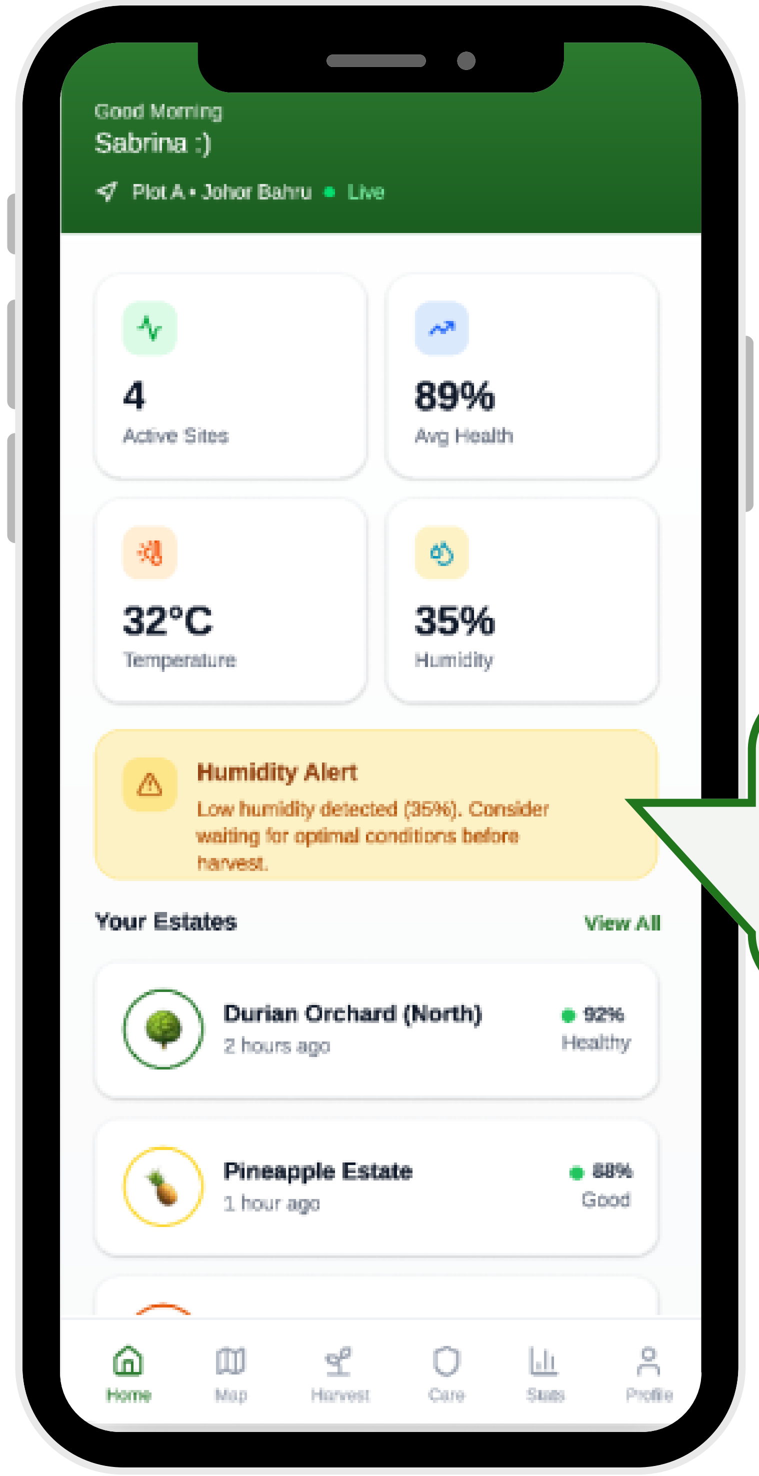


sensor detected low water level, low nutrient level and high sunlight intensity



The agent will generate AI recommendation and a task list.

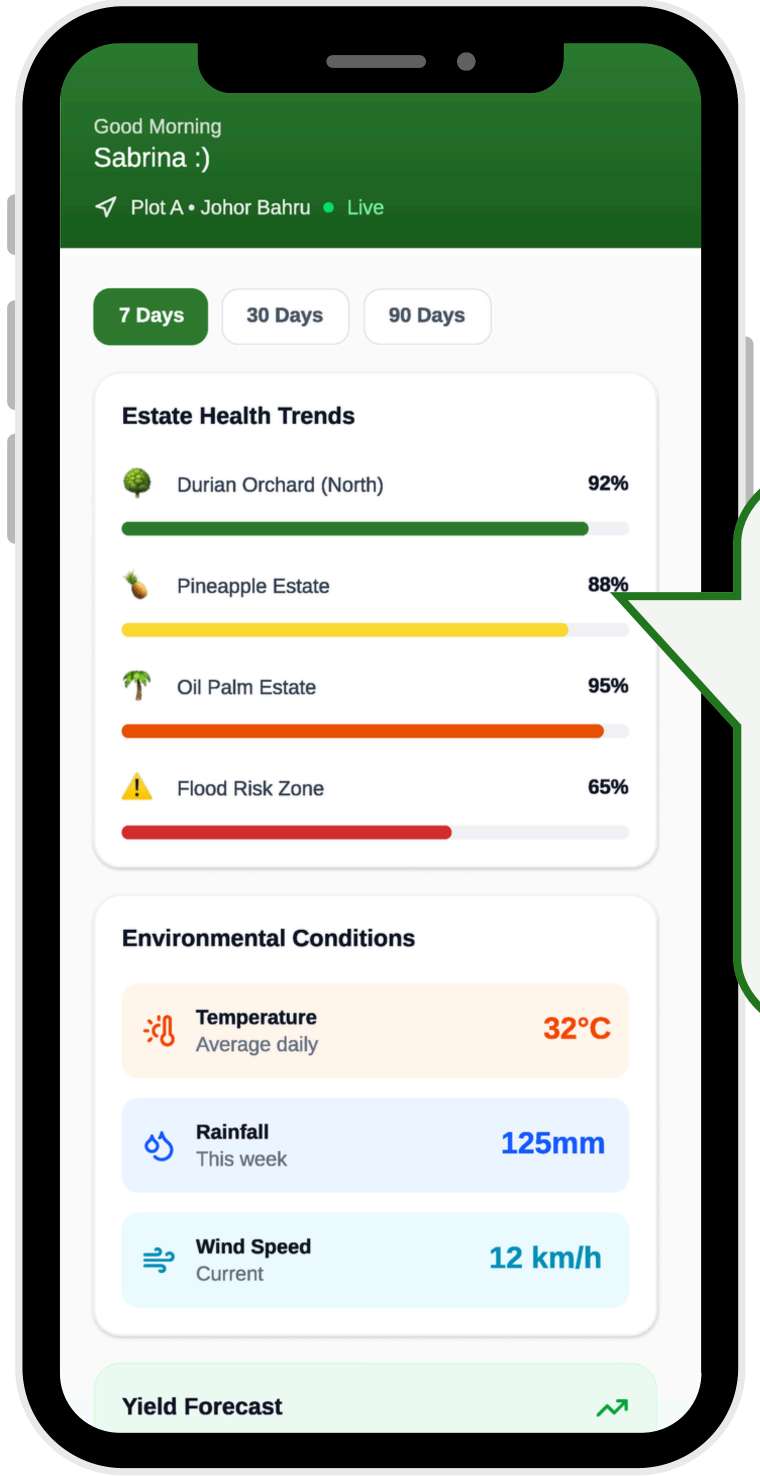
Scenario 4: Wait for Action



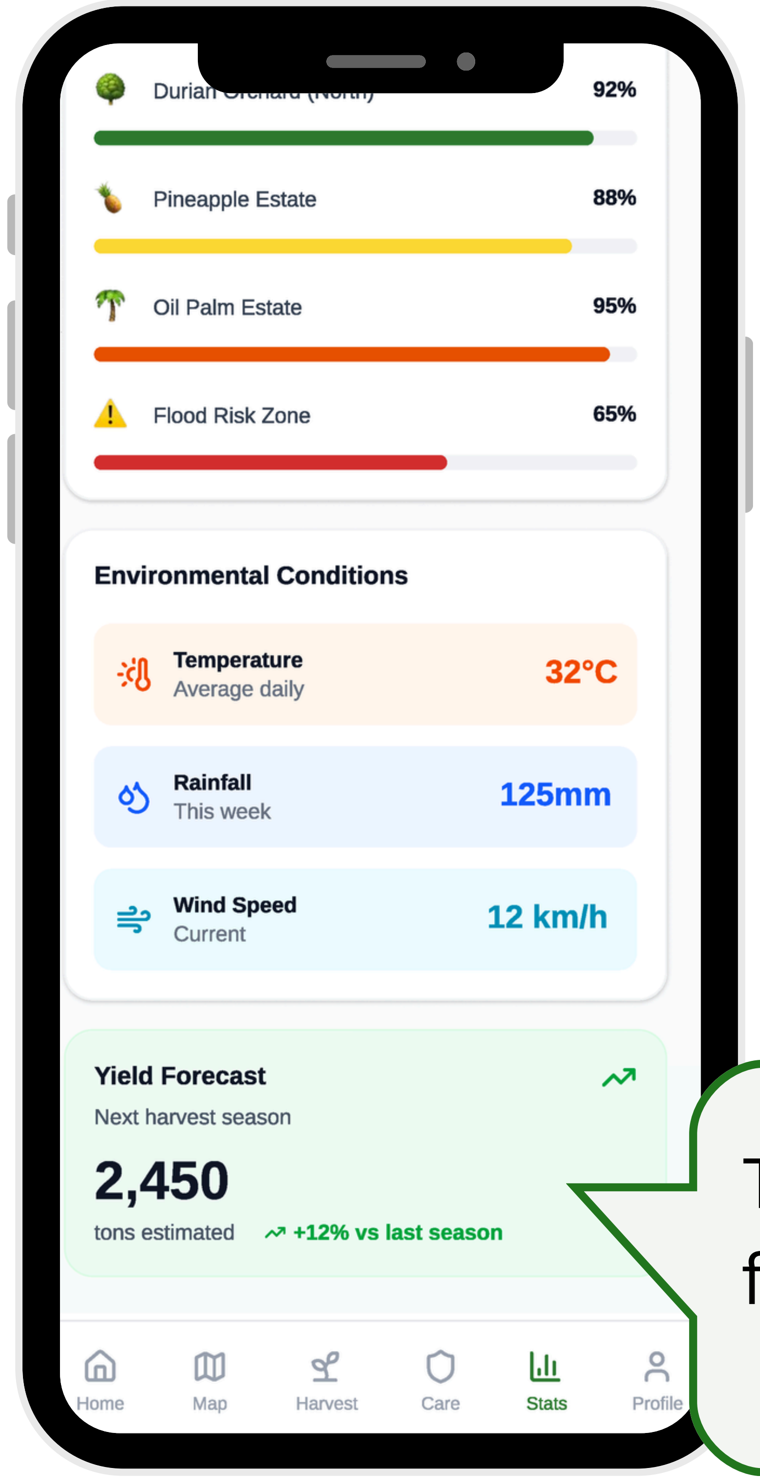
Low humidity

Wait for optimal condition to harvest.

Scenario 5: Performance Analysis & Yield Prediction



The agent will track the health trend to provide the farmer the latest status.



The agent provide yield forecast by aggregating the sensor data.